

Supplementary Material

Physics-Informed Intent-Aware Human-Robot Collaborative Transportation With Forward-Looking Adaptation

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I. IMPLEMENTATION DETAILS

We adopt an encoder-decoder architecture with a forward dynamics predictor for PIFA. At each time step, the proprioceptive observation contains 45 dimensions, consisting of leg joint positions and velocities, arm joint positions and velocities, base linear velocity, base orientation, and base angular velocity. The complete observation space is summarized in Table I.

TABLE I
PROPRIOCEPTIVE OBSERVATION USED BY PIFA.

Quantity	Description	Dim.
$\mathbf{q}$	Leg joint positions	12
$\dot{\mathbf{q}}$	Leg joint velocities	12
$\boldsymbol{\eta}$	Arm joint positions	6
$\dot{\boldsymbol{\eta}}$	Arm joint velocities	6
$\dot{\mathbf{p}}$	Base linear velocity	3
$\boldsymbol{\theta}$	Base orientation	3
$\dot{\boldsymbol{\theta}}$	Base angular velocity	3
Total		45

A history of $L = 20$ observations is sampled at 50 ms intervals, corresponding to 1.0 s of temporal context and yielding the 900-dimensional history input $\mathcal{H}_t$. The 500 Hz proprioceptive stream is temporally subsampled when constructing this history. All input features are standardized using statistics computed from the training split, and the same fixed normalization is used during deployment. The predictor internally represents the future robot configuration $\hat{\boldsymbol{\rho}}_{t+1}$ using the base pose and joint configuration. The base pose is represented as $[\Delta x, \Delta y, h, \text{roll}, \text{pitch}, \text{yaw}]$, where horizontal translation is expressed relative to the current base pose, while base height and orientation are represented in absolute coordinates. The joint output contains the 12 leg joints and 6 arm joints defined in $\boldsymbol{\rho}$.

PIFA obtains the generalized velocity and acceleration from the predicted configuration over the fixed prediction interval Δt :

$$\hat{\boldsymbol{\rho}}_{t+1} = \frac{\hat{\boldsymbol{\rho}}_{t+1} - \boldsymbol{\rho}_t}{\Delta t}, \quad \hat{\dot{\boldsymbol{\rho}}}_{t+1} = \frac{\hat{\boldsymbol{\rho}}_{t+1} - \dot{\boldsymbol{\rho}}_t}{\Delta t}. \quad (1)$$

The predicted roll-pitch-yaw rates are converted to body angular velocity before computing the corresponding angular acceleration. The whole-body dynamics quantities $\mathbf{M}$, $\mathbf{h}$, and $\mathbf{J}$ are evaluated using Pinocchio at the predicted configuration. During backpropagation, these quantities are treated as locally constant, while gradients propagate through the predicted acceleration, generalized force, and interaction wrench. The dynamics residual is normalized by the sum of the magnitudes of its left- and right-hand sides using a stop-gradient operation to avoid gradients through the normalization term.

II. TRAINING DETAILS

Training data are collected in MuJoCo using a Unitree Go1 quadruped equipped with a K1 manipulator and a virtual human partner carrying the opposite side of the payload. Each episode samples a human motion primitive, including forward and backward motion, left and right lateral motion, upward and downward motion, yaw rotation, and stopping, while the payload mass is uniformly sampled from $[0.6, 1.2]$ kg. External disturbances are introduced independently at the robot base and manipulator through randomized pushes in arbitrary directions, with force magnitudes independently sampled from $[2, 10]$ N. The model is trained end-to-end using the context-encoding loss, whole-body dynamics loss, prediction loss, and velocity loss defined in the main paper. In implementation, the learnable loss components are balanced using homoscedastic uncertainty weighting,

$$\lambda_i = e^{-s_i}, \quad (2)$$

where s_i is constrained to $[-3, 3]$. The reconstruction and KL regularization are treated as fixed regularizers with unit weight. The resulting loss weights of the best validation checkpoint are summarized in Table II.

We train the model using Adam with a learning rate of 10^{-3} and a batch size of 256. The dataset is split into training and validation sets, and the best validation checkpoint is used for evaluation.

TABLE II
LOSS COMPONENTS AND CORRESPONDING WEIGHTS USED DURING TRAINING.

Loss	Component / Expression	λ
$\mathcal{L}_{wb}$	Whole-body dynamics residual	20.22
$\mathcal{L}_{vel}$	$\ \dot{\mathbf{q}}_{t+1} - \hat{\dot{\mathbf{q}}}_{t+1}\ _2^2$	20.19
	$\ \dot{\mathbf{h}}_{t+1} - \hat{\dot{\mathbf{h}}}_{t+1}\ _2^2$	20.21
$\mathcal{L}_{pred}$	$\ \mathbf{s}_{t+1} - \hat{\mathbf{s}}_{t+1}\ _2^2$	20.22
	$\ \mathbf{Q}_{t+1} - \hat{\mathbf{Q}}_{t+1}\ _2^2$	0.93
	$\ \mathbf{b}_t - \hat{\mathbf{b}}_t\ _2^2$	20.21
	$\ \mathbf{w}_t - \hat{\mathbf{w}}_t\ _2^2$	2.63
$\mathcal{L}_{CE}$	$\ \mathcal{H}_t - \hat{\mathcal{H}}_t\ _2^2$	1.00 [†]
	$D_{KL}(q(\mathbf{z}_t \mathcal{H}_t)\ \mathcal{N}(\mathbf{0}, \mathbf{I}))$	1.00 [†]

Learnable loss components are weighted using homoscedastic uncertainty, $\lambda_i = e^{-s_i}$, with $s_i \in [-3, 3]$. [†]The $\mathcal{L}_{CE}$ components are fixed regularizers.

III. EXECUTION DETAILS

During deployment, the most recent proprioceptive observations are stored in a history buffer and passed to the context encoder to obtain the latent representation $\mathbf{z}_t$. The forward dynamics predictor then estimates the future robot state, non-intent force, human motion intent, and interaction wrench. These predictions are used to generate coordinated leg-torque and arm-position commands through the nominal whole-body controller, forward-looking adaptive compensation, and admittance-based arm controller. Algorithm 1 summarizes the complete online execution procedure.

Algorithm 1: Online Execution of PIFA

Input : Robot state history $\mathcal{H}_t$, current robot state $\mathbf{s}_t$, desired grasp pose $\mathbf{g}^d$, adaptive matrices $\mathbf{G}_t$ and $\mathbf{R}_t$
Output : Leg torque command $\boldsymbol{\tau}_t^{\text{cmd}}$ and arm joint-position command $\mathbf{a}_t^{\text{cmd}}$

- 1 Compute latent context $\mathbf{z}_t = \phi(\mathcal{H}_t)$;
- 2 Predict future robot state, non-intent force, human motion intent, and interaction wrench: $(\hat{\mathbf{s}}_{t+1}, \hat{\mathbf{Q}}_{t+1}, \hat{\mathbf{b}}_t, \hat{\mathbf{w}}_t) = f_\theta(\mathbf{s}_t, \mathbf{z}_t)$;
- 3 Compute nominal leg position and torque commands: $(\mathbf{q}_{t+1}^d, \boldsymbol{\tau}_t^d) = \pi(\mathbf{s}_t, \hat{\mathbf{b}}_t)$;
- 4 Extract $\hat{\mathbf{q}}_{t+1}$ from $\hat{\mathbf{s}}_{t+1}$ and compute the predicted joint deviation $\Delta\mathbf{q}_{t+1} = \mathbf{q}_{t+1}^d - \hat{\mathbf{q}}_{t+1}$;
- 5 Compute the joint-state and torque tracking errors $\mathbf{e}_q$ and $\mathbf{e}_\tau$;
- 6 Update the adaptive matrices $\mathbf{G}_t$ and $\mathbf{R}_t$ using the bounded-gain-forgetting adaptive laws;
- 7 Compute proactive torque compensation $\Delta\boldsymbol{\tau}_t = \mathbf{G}_t\Delta\mathbf{q}_{t+1}$;
- 8 Generate the compensated leg torque command $\boldsymbol{\tau}_t^{\text{cmd}} = \boldsymbol{\tau}_t^d + \Delta\boldsymbol{\tau}_t$;
- 9 Compute the compliant end-effector offset $\Delta\mathbf{x}_t$ from $\hat{\mathbf{w}}_t$ using the second-order admittance model;
- 10 Compute the vertical intent offset $\Delta\mathbf{x}_{z,t} = \mathbf{c} \hat{v}_{z,t} \Delta t$;
- 11 Construct the desired end-effector pose $\mathbf{x}_t^d = \mathbf{g}^d + \Delta\mathbf{x}_t + \Delta\mathbf{x}_{z,t}$;
- 12 Generate the arm joint-position command $\mathbf{a}_t^{\text{cmd}} = f_{IK}(\boldsymbol{\eta}_t, \mathbf{x}_t^d)$;
- 13 **return** $\boldsymbol{\tau}_t^{\text{cmd}}, \mathbf{a}_t^{\text{cmd}}$

The predicted interaction wrench drives a second-order admittance model about the desired grasp pose,

$$\mathbf{M}_a \ddot{\Delta\mathbf{x}}_t + \mathbf{D}_a \dot{\Delta\mathbf{x}}_t + \mathbf{K}_a \Delta\mathbf{x}_t = \hat{\mathbf{w}}_t. \quad (3)$$

We set the diagonal entries of $\mathbf{M}_a$, $\mathbf{D}_a$, and $\mathbf{K}_a$ to 1.5 kg, 58 N·s/m, and 450 N/m, respectively, for all three Cartesian axes.

To suppress drift caused by estimation noise when no physical interaction is present, the predicted base command is gated using the estimated interaction force. The gate begins to activate at $f_{\text{on}} = 0.5$ N and increases linearly over the subsequent 0.3 N range, followed by a 0.20 s low-pass filter.

The predicted next robot state is used to compute the forward-looking joint deviation $\Delta\mathbf{q}_{t+1}$ relative to the nominal locomotion command. For the BGF adaptive law [1], we use $\kappa = 1.2$, $\lambda_0 = 3.0$, $k_0 = 30.0$, and $\alpha = 20.0$, with initial values $\mathbf{R}_0 = (100/12)\mathbf{I}$ and $\mathbf{G}_0 = \mathbf{0}$. The predicted joint deviation is norm-clamped to 2.0, and $\mathbf{G}_t$ and $\mathbf{R}_t$ are updated sequentially at each control step before the resulting torque compensation is applied to the nominal leg command.

PIFA generates locomotion commands at 500 Hz, while the nominal locomotion controller [2] and BGF update law operate at 1 kHz. The estimator uses a 1.0 s history constructed from observations sampled at 50 ms intervals. All real-robot experiments are executed onboard a Unitree Go1 quadruped equipped with a K1 manipulator and an NVIDIA Jetson Orin computer.

REFERENCES

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- [2] D. Kim, J. Di Carlo, B. Katz, G. Bledt, and S. Kim, “Highly dynamic quadruped locomotion via whole-body impulse control and model predictive control,” *arXiv*, 2019.